NutriVision – AI-Based Food Image Classification

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GROUP 2

Project Overview

Core Goal

Build a complete end-to-end intelligent food analysis system tailored for Indian cuisine.

Problem Statement

- Finding each meal's caloric value takes long, often leads to mistakes
- Logging what you eat usually needs hand-entered data in most apps
- Portion size estimation from images is inaccurate
- Indian mixed meals are underrepresented in datasets

Proposed Idea – NutriVision

An AI-powered system that:

- Finding several foods within one picture of a meal
- Fraction of food measured by comparing against a standard plate size
- Maps detected food to a structured nutrition database
- Performs personalized nutrient gap analysis
- Ai Suggests Meals Based On Diet

Literature Review: Food Image Classification

Deep Learning in Food Analysis

- CNN-based models widely used for food classification
- Food-101 dataset: 101 categories, 100,000+ images
- Transfer learning with architectures like ResNet and EfficientNet improves performance significantly

Object Detection Advances

- Finding objects fast, today's setups rely on YOLO to spot many things at once. Though speed matters, accuracy stays strong across varied scenes. Because processing happens instantly, results appear without delay during operation
- Detects multiple food items in a single image
- Bounding-box detection enables identification of each food component separately
- Focusing on varied dishes works better than sorting by one category alone

Research Gap

- most data covers western food
- Indian mixed meals are underrepresented
- Most current frameworks stick to sorting things into just one category at a time
- Need adaptation for culturally diverse food types

Literature Review: Nutrition Estimation Systems

Existing Approaches

- MyFitnessPal and CalorieMama monitor calories
- Many systems require manual meal logging
- Using pictures, certain tools figure out energy content without manual input

Key Limitations

- Manual logging leads to effort and user errors
- Portion sizes often standardized (e.g., fixed 100g assumptions)
- Few visual hints in flat pictures make it hard to judge how much is really there
- Back then, people needed tools like rulers or dedicated devices to get things done

Identified Gaps

- Limited personalized nutrition gap analysis
- Weak integration between detection and dietary advice
- No strong end-to-end system combining:
- Food detection
- Portion estimation
- Structured nutrient mapping
- Intelligent dietary suggestions

Contribution of Proposed Solution

NutriVision – Key Contributions

1. Multi-Food Detection (YOLOv8 Based)

- Detects multiple food items in a single plate image
- Identifies both plate boundary and individual food items
- Designed specifically for Indian mixed meals

2. Improved Portion Size Estimation

- Uses plate-reference pixel area scaling
- Converts visual proportions into real-world weight (grams)
- No need for depth cameras or manual measurements

3. Custom Indian Nutrition Database

- Built using IFCT + USDA + custom datasets
- Includes regional Indian dishes
- Structured CSV-based extensible design

4. Personalized Nutritional Gap Analysis

- Uses Mifflin-St Jeor equation for BMR calculation
- Compares intake vs required daily nutrition
- Moves beyond simple calorie counting

5. Conversational AI Guidance

- Claude LLM generates structured dietary suggestions
- Provides meal balance feedback
- Suggests nutrient corrections and supplements

Model Architecture

Four-Stage Sequential Pipeline

Stage 1: Detection

- YOLOv8 detects food items and plate
- Outputs bounding boxes + class labels + confidence scores

Stage 2: Segmentation

- SAM (Segment Anything Model)
- Generates pixel-level masks from YOLO prompts

Stage 3: Weight Estimation

- Solid foods → Depth estimation (ZoeDepth)
- Liquid foods → Geometric bowl-volume estimation
- Weight = Volume × Density

Stage 4: Nutrition Mapping

- Maps food label + grams to database
- Scales nutrients using per-100g formula
- Output → Complete Nutritional Breakdown

Food Detection & Portion Estimation Pipeline

Step-by-Step Process

1. Upload Meal Image
2. YOLOv8 detects:
   - Plate boundary
   - Multiple food items
3. SAM generates precise segmentation masks
4. Pixel area calculated for each food
5. Plate-to-food ratio computed
6. Volume estimated
7. Weight (grams) calculated using density

Formula Used:

Scaled Weight $\propto$ (Food Pixel Area / Plate Pixel Area)

This ensures realistic portion estimation without manual input.

Nutrition Mapping Layer

Custom Nutrition Database

Stores values per 100g:

- Calories (kcal)
- Protein (g)
- Carbohydrates (g)
- Fat (g)
- Fiber (g)
- Iron (mg)
- Calcium (mg)
- Vitamins (A, B, C, D)

Nutrition Scaling Formula

- Scaled Nutrition = (Nutrition per 100g × Estimated grams) / 100

Data Sources

- USDA FoodData Central
- Indian Food Composition Tables (IFCT)
- Custom Indian datasets

Ensures culturally accurate nutrient estimation.

Conversational Suggestion Layer

AI-Powered Personalized Guidance

Uses Claude LLM API

Input to LLM:

- User profile:
  - Age
  - Height
  - Weight
  - Activity level
- Detected foods
- Estimated nutrients
- Nutrient gap analysis

Output:

1. Meal balance feedback
2. Suggestions to correct deficiencies
3. Supplement guidance (if required)

No fine-tuning

No external RAG

Prompt-engineered structured suggestions

Dataset Availability

Food Detection Datasets

- Food-101: 101K images, 101 categories (benchmark pretraining)
- Indian Food Images Dataset: 4,000 images, 80 Indian dishes
- Custom Dataset: Mixed Indian meals, annotated using Roboflow (bounding boxes)

Nutrition Datasets

- Indian Food 101: 250 dishes (ingredients & origin)
- Indian Food Nutritional Values (2025): Calories, macros, micronutrients
- Extendable custom nutrition database planned

Computation Feasibility

- Training on Google Colab Pro (NVIDIA T4 GPU)
- Model: YOLOv8 (Transfer Learning approach)
- Fine-tuning final layers only
- Training time: 2–4 hours
- Batch size optimized based on GPU memory
- Post-training pipeline requires minimal computation

Evaluation Metrics

Food Detection Evaluation

- Mean Average Precision (mAP) – overall detection performance
- Precision – correctness of detected items
- Recall – ability to detect all items
- F1 Score – balance between precision & recall

Portion Size Estimation

- Plate-to-food pixel area ratio method
- Ground truth dataset with actual weights
- Mean Absolute Error (MAE) for evaluation

Gap Analysis & Suggestion Evaluation

- BMR calculated using Mifflin-St Jeor equation
- Compared with WHO nutritional intake guidelines
- Ensures no unsafe recommendations
- Relevance Score via human evaluation (normalized rating scale)

Conclusion

- NutriVision integrates food detection, portion estimation, and nutritional analysis into a unified pipeline.
- Combines public + custom Indian cuisine datasets for improved accuracy and cultural relevance.
- Uses YOLOv8 with transfer learning for efficient and scalable model training.
- Ensures reliability through robust evaluation metrics (mAP, MAE, relevance scoring).
- Provides safe, guideline-based nutrition suggestions aligned with WHO standards.

Outcome:

A scalable, accurate, and culturally adapted AI-driven nutrition analysis system for Indian meals.